

CSA0306-DATA STRUCTURES

BANKING SYSTEM USING HASH TABLE

INSTANT CUSTOMER ACCOUNT RETRIEVAL

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Problem Statement

- **Problem**
- A banking system must retrieve customer account details instantly using account numbers.
- **Requirements**
- Fast searching
- Easy insertion
- Easy updating
- Efficient storage

Suggested Data Structure

- Hash Table (Hash Map)
- A Hash Table stores data as (Key, Value) pairs.
- Key → Account Number
- Value → Customer Details

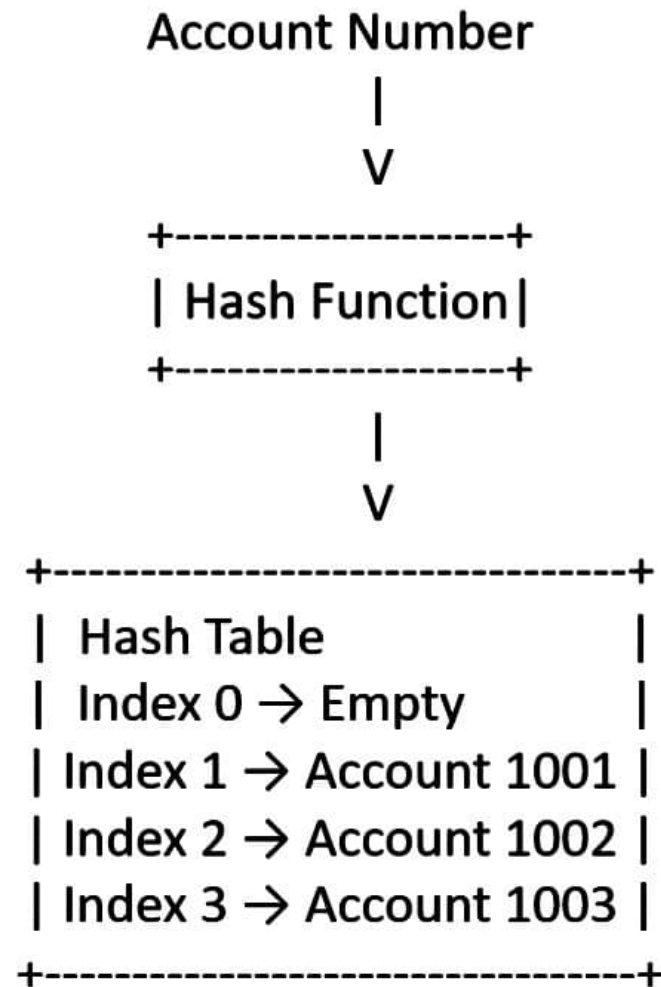
Example:

Account No	Customer
1001	Arun
1002	Priya
1003	Rahul

Working

- **Hash Function**
- Hash(Account Number)
- Example
- 1001 → Index 1
- 1002 → Index 2
- 1003 → Index 3
- The account number is converted into an array index, allowing direct access.

Diagram



Time Complexity

Operation	Complexity
Search	$O(1)$
Insert	$O(1)$
Delete	$O(1)$

Worst Case (Collision): $O(n)$

Average Case: $O(1)$

Advantages

- Very fast searching
- Instant account retrieval
- Easy insertion
- Easy deletion
- Efficient memory usage
- Suitable for banking applications

Conclusion

Hash Table is the best data structure for banking systems. It provides constant-time ($O(1)$) search.

Customer details can be retrieved instantly using account numbers.

It improves system performance and efficiency

Thank You!



SIMATS ENGINEERING

ASSESSMENT TOOL 3 - Technical Presentation

COURSE NAME: Data Structure
Assessment Tool Weightage: 30%

COURSE CODE: CSA03

COURSE OUTCOME COVERED

CO1: Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity. (BL3, BL4)

Questions: EACH QUESTIONS CARRIES : 10 MARKS

Duration : ONE HOUR

S.No	List of Questions
1	A student management system needs to store the marks of 1000 students efficiently. Apply the concepts of primitive and non-primitive data structures to select an appropriate structure for storing and managing the data and prepare a technical presentation justifying your choice based on the efficiency of data structure.
2	Apply a suitable data structure to implement an LRU cache system. Prepare a technical presentation explaining how it improves performance.
3	Apply suitable data structure for the application that encourages passengers to book tickets online and how the requests are processed in order of arrival. Prepare a technical presentation justifying its efficiency using complexity analysis.
4	Choose a suitable linear or non-linear data structure for a navigation system that stores city connections and finds shortest routes between locations. Prepare a technical presentation explaining why it is appropriate.
5	A web browser supports navigating backward through previously visited pages. Recommend a suitable ADT and prepare a technical presentation explaining how it improves performance efficiency.
6	A digital library allows searching books using ISBN numbers quickly among 50,000 records. Select the most suitable data structure and Prepare a technical presentation justifying using time complexity comparison.
7	A social networking platform stores friend relationships between users. Recommend a suitable representation structure and Prepare a technical presentation explaining why it supports efficient relationship mapping.
8	Customer service requests are processed in the order they arrive. Identify the suitable ADT and Prepare a technical presentation explaining why it ensures fair request processing.
9	A programmer uses recursion to compute factorial values. Explain how recursion internally uses a data structure and analyze its space complexity implications. Prepare a technical presentation.



10	An institute frequently sorts student marks before publishing rank lists. Suggest a suitable data structure for storing the records and justify your selection based on sorting efficiency considerations. Prepare a technical presentation.
11	A hospital emergency ward treats patients based on severity rather than arrival order. Suggest a suitable data structure for managing patients and prepare a technical presentation explaining your choice.
12	A music streaming application maintains a playlist where songs can be added, removed, or reordered dynamically. Recommend a suitable data structure and justify based on flexibility and efficiency. Prepare a technical Presentation.
13	A dictionary application needs to quickly search whether a word exists in its database. Suggest an appropriate data structure and justify based on search performance. Prepare a technical Presentation.
14	An operating system manages multiple processes waiting for CPU allocation in a time-sharing environment. Recommend an appropriate data structure and justify based on time and space efficiency.
15	An e-commerce website stores product information and frequently updates stock availability. Recommend a suitable ADT and prepare a technical presentation explaining how it improves performance efficiency.
16	A printer processes multiple print requests in the order they are received. Recommend an appropriate data structure and prepare the presentation explaining your choice.
17	A spell-checking application verifies whether typed words exist in its dictionary database. Suggest a suitable data structure and justify based on search speed.
18	A metro rail network maps stations and routes to determine reachable stations from a source station. Suggest a suitable data structure for representation and justify your answer.
19	A compiler checks whether parentheses in a program are balanced during syntax analysis. Identify the suitable data structure and justify based on operational requirements.
20	An online leaderboard system continuously updates player rankings after each match. Suggest an efficient data structure and justify based on sorting and update efficiency.
21	A task scheduling system assigns jobs based on execution priority levels. Suggest a suitable data structure and justify performance efficiency.
22	A maze-solving algorithm explores multiple paths recursively. Recommend the underlying structure and justify its efficiency.
23	A contact management application frequently inserts and deletes user records dynamically. Suggest a suitable data structure and justify your selection.
24	A banking system retrieves customer account details using account numbers instantly. Suggest an



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Code of Conduct:

I Mounnila S P(192511188) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness